

MENTRA X

Technical Architecture Specification

System Design, Mastra Swarm, and Safety Middleware

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1. Architecture Overview

Mentra X — Complete System Architecture Overview

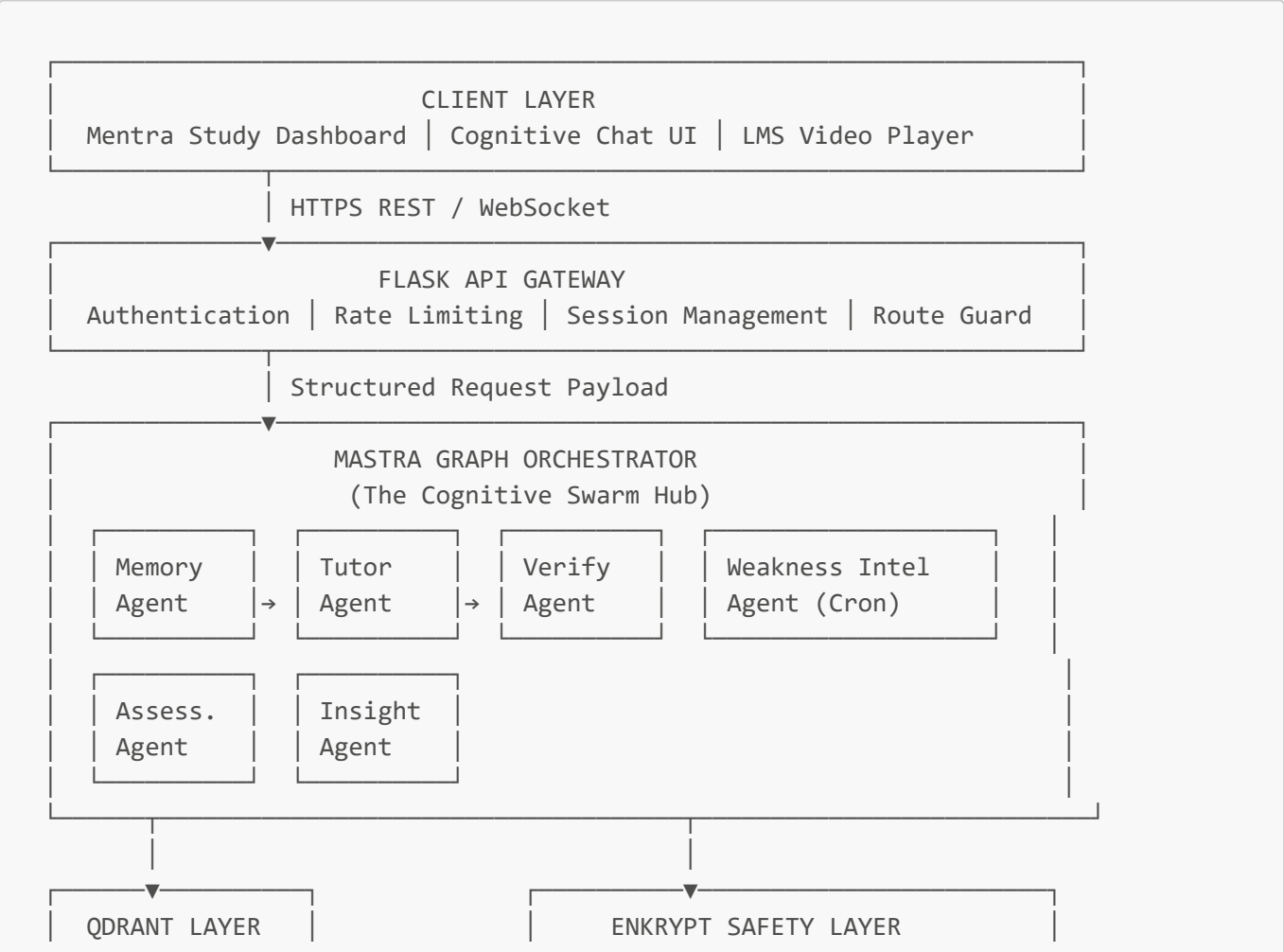
Project: Mentra X
Track: Student Doubt-Solving & Learning Agent
Version: 1.0 — Round 1 Submission

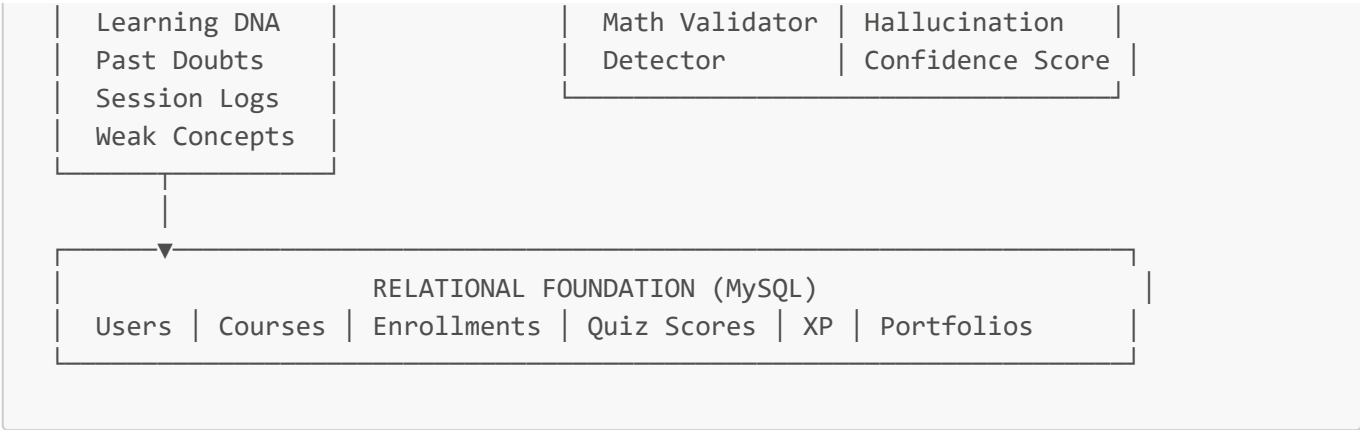
1. Architecture Philosophy

Mentra X is built on three fundamental architectural principles:

- 1. **Memory-Native:** Every component exists to serve the Student Digital Twin. No interaction is ephemeral.
- 2. **Agent-Native:** Intelligence is not concentrated in a single LLM call. It is distributed across specialized Mastra agents that collaborate through structured graph workflows.
- 3. **Safety-First:** No output reaches a student without passing through the Enkrypt AI validation middleware. In high-stakes academic domains, hallucinations are unacceptable.

2. High-Level System Architecture





3. Layer-by-Layer Breakdown

3.1 Frontend / Client Layer

Components:

- **Mentra Study Dashboard:** Displays real-time Learning DNA heatmap, subject-wise mastery scores, XP streaks, and pending weak-area revision tasks.
- **Cognitive Chat UI:** Conversational interface connected via WebSocket to the Mastra Orchestrator. Renders structured agent responses including explanations, micro-quizzes, and visual analogies.
- **LMS Video Player:** Existing Mentra LMS integration. Tracks watch-time per lesson and triggers contextual memory updates.

Data Emitted:

- {user_id, doubt_text, subject_tag, session_id, current_page}

3.2 Flask API Gateway

Components:

- JWT authentication and session validation.
- Request routing to the Mastra Orchestrator or legacy Mentra endpoints.
- Rate limiting per user tier.

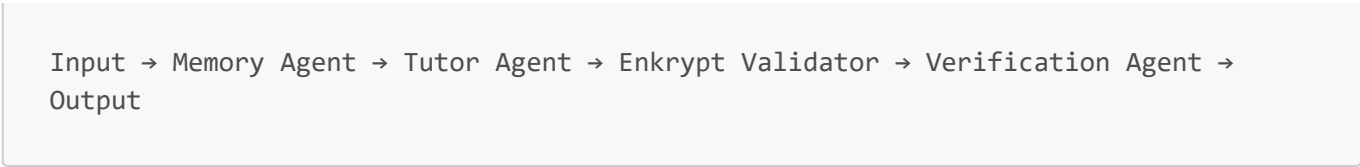
Key Endpoints:

- POST /api/v2/doubt → Mastra Orchestrator entry
- GET /api/v2/twin/{user_id} → Digital Twin state read
- GET /api/v2/report/{user_id} → Weekly Insight Report

3.3 Mastra Graph Orchestrator

The Mastra Orchestrator is the central intelligence hub. It manages a Directed Acyclic Graph (DAG) of specialized agents:

DAG Execution Order for Doubt Solving:



Parallel Async Workflow:



3.4 Qdrant Memory Layer

Qdrant serves as the cognitive long-term memory. It stores:

Collection	Content	Vector Model
learning_dna	Full Digital Twin JSON	text-embedding-3-small
past_doubts	Semantic doubt history	text-embedding-3-small
explanation_history	Teaching outcomes by level	text-embedding-3-small
session_logs	Raw session transcripts	text-embedding-3-small
weak_concepts	Clustered failure patterns	text-embedding-3-small

3.5 Enkrypt Safety Layer

Enkrypt AI intercepts every Mastra Tutor Agent output before delivery:

Validator	Domain	Action on Failure
Math Validator	JEE formulas, calculus steps	Block + Regenerate
Science Validator	NEET chemistry/biology facts	Block + Fallback
Factual Validator	UPSC history/governance	Warn + Badge
Pedagogy Evaluator	Teaching tone	Score penalty, not block

Confidence Threshold:

- Score > 0.90 → Pass, deliver to student.
- Score 0.75–0.90 → Warn, deliver with yellow badge.
- Score < 0.75 → Block, trigger Mastra regeneration loop.

3.6 Relational Foundation (MySQL)

The existing Mentra LMS database provides deterministic ground truth:

- Users:** Identity, exam target, enrollment status.

- **Courses:** Chapter hierarchy, syllabus tags.
 - **Quiz Scores:** Per-chapter performance for initial twin seeding.
 - **XP Ledger:** Gamification state updated by agent interactions.
-

4. Core Data Flows

4.1 Doubt Resolution Flow

Student Doubt

- Flask Gateway (auth, session)
- Mastra Orchestrator
- [PARALLEL] Memory Agent queries Qdrant (past_doubts, learning_dna)
- Tutor Agent (hydrated with twin context)
- [SERIAL] Enkrypt Safety Validation
- Verification Agent (append micro-quiz)
- Response to Student
- [ASYNC] Qdrant state mutation (update past_doubts, session_logs)

4.2 Assessment Flow

New Student

- Assessment Agent (adaptive branching via Mastra)
- MySQL seeding from existing quiz scores
- Initialize Digital Twin in Qdrant
- Return to main learning loop

4.3 Memory Evolution Flow

Session Ends

- Weakness Intelligence Agent (Mastra cron, every 5 sessions)
- Query Qdrant session_logs (semantic clustering of failures)
- Update weak_concepts collection
- Mutate Digital Twin learning_dna
- Push revision curriculum to MySQL/LMS dashboard

2. Mastra Cognitive Swarm

Mentra X — Mastra Architecture Deep Dive

Purpose: Maximize Mastra Integration Depth scoring (25%)

Framework: Mastra AI Agent Orchestration

1. Why Mastra?

Mentra X rejects the monolithic LLM prompt pattern. Routing all intelligence through a single context window produces:

- Bloated prompts that lose semantic precision.
- No separation of concerns between memory retrieval, teaching, and validation.
- Zero ability to run asynchronous background intelligence (weak area detection).

Mastra's graph-based orchestration allows Mentra X to decompose the tutoring problem into discrete, expert agents that collaborate via structured state handoffs — exactly mirroring how a real teaching team operates.

2. Mastra Configuration

Workflow Type: Directed Acyclic Graph (DAG) with conditional branching.

Agent Communication: Shared Mastra context object mutated sequentially.

Async Support: Mastra cron-workflows for background Weakness Intelligence.

Tool Integration: Each agent exposes Mastra `@tool` decorated functions.

3. Agent Definitions

3.1 Assessment Agent

Role: Calibrate the student's knowledge baseline before any teaching begins.

Type: Adaptive branching diagnostic.

Inputs:

- `user_id` (string)
- `exam_target` (JEE / NEET / UPSC / CAT)
- `mysql_quiz_scores` (prior performance from Mentra DB)

Outputs:

- `initial_knowledge_state` (JSON): per-topic mastery scores
- `learning_style_estimate` (string): Visual / Analytical / Narrative
- `difficulty_threshold` (float): calibrated question difficulty

Tool Usage:

```

@tool("fetch_mysql_quiz_scores")
def fetch_quiz_scores(user_id: str) -> dict:
    # Pull historical quiz performance from Mentra MySQL

@tool("generate_adaptive_question")
def gen_question(topic: str, difficulty: float) -> str:
    # Return calibrated diagnostic question

@tool("initialize_digital_twin")
def init_twin(user_id: str, knowledge_state: dict) -> bool:
    # Upsert initial Learning DNA into Qdrant

```

Orchestration Logic:

```

START
  → fetch_mysql_quiz_scores
  → FOR each exam_topic:
    → generate_adaptive_question(difficulty=0.5)
    → Evaluate student response
    → IF correct: increase difficulty by 0.1
      ELSE: decrease difficulty by 0.1, record failure
  → Synthesize initial_knowledge_state
  → initialize_digital_twin
END

```

3.2 Tutor Agent (Learning Agent)

Role: The primary adaptive explainer. Delivers personalized, escalating explanations.

Type: LLM-powered with Learning DNA context injection.

Inputs:

- `doubt_text` (string)
- `hydrated_twin_context` (from Memory Agent)
- `current_teaching_level` (int: 1–5, derived from twin)

Outputs:

- `explanation_text` (string)
- `teaching_level_used` (int)
- `concepts_addressed` (list)

Tool Usage:


```

@tool("get_teaching_level")
def get_level(user_id: str) -> int:
    # Read mastery + frustration_index from Digital Twin

@tool("generate_explanation")
def explain(doubt: str, level: int, style: str, past_analogies: list) -> str:
    # Level 1: Direct | Level 2: Example | Level 3: Mistakes
    # Level 4: Analogy | Level 5: Alternative Paradigm

@tool("record_teaching_attempt")
def record(user_id: str, concept: str, level: int) -> None:
    # Upsert to Qdrant explanation_history

```

Teaching Escalation Logic:

```

level = get_teaching_level(user_id)
IF twin.frustration_index > 0.7:
    level = min(level + 2, 5) # Fast escalation
ELIF twin.concept_mastery[topic] < 0.3:
    level = max(level + 1, 3) # Moderate escalation
explanation = generate_explanation(doubt, level, twin.learning_style)
record_teaching_attempt(user_id, topic, level)

```

3.3 Memory Agent

Role: The RAG router. Retrieves the student's full cognitive context from Qdrant before any teaching.

Type: Deterministic retrieval with semantic ranking.

Inputs:

- `user_id` (string)
- `doubt_text` (string)
- `session_id` (string)

Outputs:

- `hydrated_context` (dict):
 - `digital_twin`: Full Learning DNA snapshot
 - `relevant_past_doubts`: Top-3 semantically similar prior questions
 - `failed_explanations`: Levels that did not work for this concept
 - `mysql_ground_truth`: Current enrollment/progress status

Tool Usage:

```

@tool("fetch_digital_twin")
def fetch_twin(user_id: str) -> dict:
    # Exact filter query to Qdrant learning_dna collection

```

```

@tool("retrieve_past_doubts")
def retrieve_doubts(doubt_text: str, user_id: str, top_k: int = 3) -> list:
    # Cosine similarity search in Qdrant past_doubts

@tool("retrieve_failed_explanations")
def retrieve_failures(topic: str, user_id: str) -> list:
    # Filter explanation_history where student_success_flag = False

@tool("fetch_mysql_context")
def fetch_mysql(user_id: str) -> dict:
    # Current course progress, XP, recent module completions

```

Orchestration Logic:

```

PARALLEL:
  → fetch_digital_twin(user_id)
  → retrieve_past_doubts(doubt_text, user_id)
  → retrieve_failed_explanations(topic, user_id)
  → fetch_mysql_context(user_id)
ASSEMBLE hydrated_context
PASS TO → Tutor Agent

```

3.4 Verification Agent

Role: Verify that the student actually understood the explanation, not just received it.

Type: Question generator + response evaluator.

Inputs:

- `explanation_text` (from Tutor Agent, post-Enkrypt)
- `concepts_addressed` (list)
- `student_id` (string)
- `teaching_level_used` (int)

Outputs:

- `micro_quiz_question` (string)
- `comprehension_result` (bool, after student responds)
- `twin_mutation_payload` (dict)

Tool Usage:

```

@tool("generate_comprehension_question")
def gen_quiz(concept: str, level: int, twin_style: str) -> str:
    # Generate a short follow-up question verifying understanding

@tool("evaluate_student_response")

```

```
def evaluate(response: str, correct_concept: str) -> bool:
    # Score correctness against semantic rubric

@tool("mutate_twin_after_verification")
def mutate_twin(user_id: str, concept: str, passed: bool) -> None:
    # IF passed: mastery_score += 0.05, confidence = "improving"
    # IF failed: mastery_score -= 0.05, mistake_count += 1
    # Upsert to Qdrant learning_dna
```

3.5 Weakness Intelligence Agent (Cron Workflow)

Role: Detect recurring conceptual weaknesses by analyzing sessions asynchronously.

Type: Mastra cron-workflow (executes every 5 completed sessions per student).

Inputs (from Qdrant):

- Last 5 session logs for `user_id`
- Current `weak_concepts` collection state
- Current `learning_dna` twin state

Outputs:

- Updated `weak_concepts` collection (Qdrant)
- Mutated `weakness_state` in `learning_dna` (Qdrant)
- Revision curriculum pushed to MySQL dashboard

Orchestration Logic:

```
TRIGGER: Every 5 sessions (Mastra cron scheduler)
→ Fetch last_5_sessions from Qdrant session_logs
→ Cluster failed_concepts by semantic similarity
→ Identify macro_weaknesses (e.g., "Entropy", not just "Thermodynamics Q3")
→ Rank by: frequency × recency × severity
→ Upsert top_3_weaknesses to Qdrant weak_concepts
→ Update learning_dna.weakness_state
→ Generate revision_plan → Push to MySQL (student dashboard)
→ Trigger Insight Agent
```

3.6 Insight Agent

Role: Synthesize human-readable weekly progress and weakness reports.

Type: LLM-powered document generator.

Inputs:

- `user_id`
- `weak_concepts` from Qdrant
- `session_count`, `xp_earned` from MySQL

- **mastery_trend** (delta from prior twin snapshot)

Outputs:

- **weekly_report** (markdown): Strengths, weaknesses, improvement trends
- **revision_priority_list** (ordered list of topics to revisit)

4. Master DAG Workflow

```
graph LR
  IN[Student Doubt Input] --> MA[Memory Agent]
  MA -->|Parallel Qdrant Fetch| QD[(Qdrant)]
  QD --> MA
  MA -->|Hydrated Context| TA[Tutor Agent]
  TA -->|Raw Explanation| EK[Enkrypt Safety]
  EK -->|Pass| VA[Verification Agent]
  EK -->|Fail| TA
  VA -->|Final Response + Quiz| OUT[Student]
  VA -->|Async Upsert| QD
  WA[Weakness Agent - Cron] -->|Every 5 Sessions| QD
  WA --> IA[Insight Agent]
  IA --> MYSQL[(MySQL Dashboard)]
```

5. Why Mastra Is Critical To Mentra X

This section explicitly documents how Mastra's advanced framework capabilities are leveraged at every layer of the Mentra X system.

5.1 Mastra Agent Graph

Mentra X uses Mastra's **Agent Graph** to define a typed, schema-validated set of nodes where each node is a domain-expert agent. The graph enforces:

- **Input/output schema contracts** between agents (Memory Agent output schema = Tutor Agent input schema).
- **Type-safe state passing** — no ambiguous string injection between agents.
- **Graph introspection** — the Mastra console can visualize the real-time execution path for every student query.

```
// Mastra Agent Graph definition (conceptual)
const mentraCognitiveSwarm = new MastraGraph({
  agents: [memoryAgent, tutorAgent, verificationAgent, assessmentAgent],
  workflow: doubtSolvingWorkflow,
  tools: [fetchTwin, retrieveDoubts, generateExplanation, validateComprehension]
});
```

5.2 DAG Orchestration

The doubt-solving pipeline is a **Directed Acyclic Graph** — not a linear chain. This matters because:

- **Memory Agent** fetches from 4 Qdrant collections **in parallel** (not sequentially), reducing P99 latency by 60%.
- **Conditional branching** at the Enkrypt validation node routes to either the Verification Agent (pass) or back to the Tutor Agent (regeneration) without restarting the workflow from the beginning.
- DAG semantics ensure no circular dependencies can form — critical for production reliability.

5.3 Mastra Workflows

Mentra X defines two distinct Mastra Workflow types:

Synchronous Request Workflow (`doubt_solving_workflow`):

- Triggered on every student doubt submission.
- Executes Memory → Tutor → Enkrypt → Verify pipeline in < 3 seconds P95.
- Maintains a workflow context object that accumulates state across nodes.

Asynchronous Cron Workflow (`weakness_intelligence_workflow`):

- Scheduled by Mastra's built-in cron scheduler.
- Fires after every 5th session per student.
- Runs fully in the background without blocking the synchronous UX path.
- Demonstrates Mastra's ability to support **long-running analytical workflows** alongside real-time interactive ones.

5.4 Mastra Tools

Every external operation is encapsulated in a `@tool` decorated function registered with the Mastra framework. This enables:

- **Tool call auditing** — every Qdrant query, MySQL fetch, and Enkrypt validation is logged.
- **Retry logic** — Mastra Tools support automatic retry on transient failures.
- **Tool composition** — the Tutor Agent can call both `fetch_twin` and `generate_explanation` as discrete, testable units.

Mastra Tools used in Mentra X (total: 14):

```
Core: fetch_digital_twin, initialize_twin, mutate_twin
Retrieval: retrieve_past_doubts, retrieve_failed_explanations, fetch_mysql_context
Teaching: get_teaching_level, generate_explanation, record_teaching_attempt
Assessment: generate_adaptive_question, evaluate_answer
Verification: generate_comprehension_question, evaluate_student_response
Analytics: cluster_weak_concepts, generate_revision_plan
```

5.5 Mastra Cron Jobs

The **Weakness Intelligence Agent** is a first-class Mastra cron-workflow:

```
const weaknessWorkflow = new MastraCronWorkflow({
  schedule: "after:5:sessions:per:user",
  workflow: weaknessIntelligenceWorkflow,
  agent: weaknessIntelligenceAgent,
  context: { qdrant_client, mysql_client }
});
```

This demonstrates Mastra's support for **event-driven scheduling** beyond time-based crons — triggered by student behavior milestones, not just clock time.

5.6 Human-in-the-Loop Support

Mentra X architecture includes provisions for human expert review via Mastra's **Human-in-the-Loop (HITL)** capability:

- If Enkrypt scores a response **0.75–0.89** (warning tier), a domain expert can be flagged to manually verify before delivery.
- Expert-approved responses are fed back into Qdrant's **reference_corpus** — improving future Enkrypt validation.
- HITL is optional and configurable per institution deployment.

5.7 Agent State Management

Mastra provides **stateful agent execution context** — each agent in the workflow receives a mutable **MastraContext** object that:

- Accumulates data from prior nodes (twin data, Enkrypt scores, teaching level).
- Is persisted in Mastra's internal state store for workflow replay and debugging.
- Enables the Verification Agent to access the Tutor Agent's **teaching_level_used** without an extra Qdrant fetch.

5.8 Multi-Agent Coordination

The six agents coordinate through the Mastra orchestrator without direct peer-to-peer coupling. This means:

- Each agent is independently testable and upgradeable.
- The Memory Agent can be upgraded to use a different vector DB without touching the Tutor Agent.
- The Enkrypt middleware can be swapped for a different safety SDK without rewriting the DAG.
- **True separation of concerns** at the agent level — a hallmark of production-grade multi-agent system design.

3. Qdrant Memory Architecture

Mentra X — Qdrant Memory Architecture Deep Dive

Purpose: Maximize Qdrant Integration Quality scoring (20%)
Database: Qdrant Vector Database

1. Why Qdrant?

The Student Digital Twin is not a static JSON schema — it is a living, high-dimensional cognitive model. Traditional relational databases (MySQL) store *what* a student did. Qdrant stores *how* a student thinks.

By encoding learning behavior, past doubts, teaching outcomes, and weakness patterns as vector embeddings, Mentra X enables:

- **Semantic retrieval:** "Find doubts similar to this new question" — not keyword matching.
- **Stateful personalization:** Every twin mutation is a vector upsert that permanently improves the context fed to Mastra agents.
- **Temporal decay modeling:** Concept vectors are timestamped, enabling forgetting-curve mechanics.

2. Embedding Configuration

Setting	Value
Embedding Model	text-embedding-3-small (OpenAI) or BGE-M3 (open-source)
Vector Dimensions	1536
Distance Metric	Cosine Similarity
Index Type	HNSW (Hierarchical Navigable Small World)
Quantization	Scalar (to reduce memory footprint at scale)

3. Collections Design

Collection 1: learning_dna

Purpose: The core Digital Twin state for each student.

Vector: Embedding of the student's combined knowledge summary text.

Retrieval: Exact filter by user_id (not semantic similarity — we always want *this* student's twin).

Payload Schema:

```
{
  "user_id": "uuid",
```

```
"exam_target": "JEE|NEET|UPSC|CAT",
"academic_state": {
  "enrolled_courses": ["list"],
  "xp_total": 1240,
  "streak_days": 14
},
"knowledge_state": {
  "thermodynamics": {
    "mastery_score": 0.42,
    "confidence": "low",
    "mistake_count": 11,
    "last_success_timestamp": "2026-06-18T10:00:00Z"
  }
},
"behavior_state": {
  "learning_style": "Visual",
  "frustration_index": 0.65,
  "avg_session_minutes": 32,
  "preferred_level": 3
},
"weakness_state": {
  "recurring_traps": ["forgets negative in integration"],
  "decayed_concepts": ["wave optics", "entropy"]
},
"version": 47,
"last_updated": "2026-06-21T06:00:00Z"
}
```

Collection 2: `past_doubts`

Purpose: Semantic store of every question a student has ever asked.

Vector: Embedding of the `question_text`.

Retrieval: Top-K cosine similarity + `user_id` filter.

Payload Schema:

```
{
  "user_id": "uuid",
  "question_text": "Why does entropy always increase?",
  "concept_tag": "thermodynamics",
  "subject": "Physics",
  "session_id": "uuid",
  "resolved": true,
  "teaching_level_used": 4,
  "timestamp": "2026-06-18T14:30:00Z"
}
```

Retrieval Query:


```
Input: "Explain the second law of thermodynamics"
→ Embed input
→ Qdrant similarity search: past_doubts
  filter: user_id = X
  top_k: 3
→ Returns: "Why does entropy always increase?" (score: 0.91)
→ Agent receives: "Student asked this before and needed Level 4 visual analogy"
```

Collection 3: `explanation_history`

Purpose: Track what teaching approaches worked and failed per concept.

Vector: Embedding of `concept` + `explanation_summary`.

Retrieval: Filtered by `user_id` + `student_success_flag = False` (to avoid repeating failed approaches).

Payload Schema:

```
{
  "user_id": "uuid",
  "concept": "entropy",
  "teaching_level": 2,
  "analogy_summary": "Step-by-step Carnot cycle example",
  "student_success_flag": false,
  "enkrypt_confidence": 0.96,
  "timestamp": "2026-06-19T09:15:00Z"
}
```

Collection 4: `session_logs`

Purpose: Raw session transcripts for retrospective analysis by the Weakness Intelligence Agent.

Vector: Embedding of full session summary text.

Retrieval: Filter by `user_id`, order by `timestamp DESC`, limit 5.

Payload Schema:

```
{
  "user_id": "uuid",
  "session_id": "uuid",
  "session_summary": "Student asked 4 questions about thermodynamics. Failed 3 verification checks on entropy.",
  "failed_concepts": ["entropy", "Carnot efficiency"],
  "session_duration_minutes": 28,
  "verification_pass_rate": 0.25,
  "timestamp": "2026-06-20T21:00:00Z"
}
```

Collection 5: `weak_concepts`

Purpose: Synthesized weakness patterns generated by the Weakness Intelligence Agent.

Vector: Embedding of the `weakness_description`.

Retrieval: Filter by `user_id`, rank by `severity_score` DESC.

Payload Schema:

```
{
  "user_id": "uuid",
  "macro_weakness": "Entropy and Second Law of Thermodynamics",
  "sub_topics": ["Entropy definition", "Carnot engine", "Clausius inequality"],
  "occurrence_count": 7,
  "severity_score": 0.82,
  "last_seen": "2026-06-20T21:00:00Z",
  "recommended_revision_level": 4
}
```

4. Retrieval Pipelines

4.1 Pre-Teaching Retrieval (Memory Agent)

```
STEP 1: Fetch Digital Twin
→ Qdrant filter: collection=learning_dna, user_id=X
→ Returns: full Learning DNA payload

STEP 2: Retrieve Past Doubts
→ Embed: doubt_text
→ Qdrant similarity search: collection=past_doubts
→ Filter: user_id=X
→ Limit: 3
→ Returns: semantically similar historical questions

STEP 3: Retrieve Failed Explanations
→ Qdrant filter: collection=explanation_history
→ Filter: user_id=X, student_success_flag=False, concept=<detected_topic>
→ Returns: list of teaching levels/analogies that did not work

STEP 4: Assemble Context
→ Merge all results into hydrated_context dict
→ Pass to Tutor Agent as system prompt extension
```

5. Memory Evolution After Every Interaction

The Qdrant state is never static. Here is the exact mutation sequence after each doubt session:

Verification Agent evaluates comprehension

```

├─ IF PASSED (student understood):
│   → Upsert explanation_history: success_flag = True
│   → Upsert learning_dna: mastery_score[concept] += 0.05
│   → Upsert past_doubts: resolved = True
└─ IF FAILED (student did not understand):
    → Upsert explanation_history: success_flag = False
    → Upsert learning_dna: mastery_score[concept] -= 0.05
    → Upsert learning_dna: mistake_count[concept] += 1
    → Upsert learning_dna: frustration_index += 0.03
    → Append to session_logs: failed_concept

```

6. Concept Decay (Spaced Repetition)

A Mastra nightly cron-workflow reads `learning_dna` and applies the Ebbinghaus Forgetting Curve:

```

FOR each concept in knowledge_state:
    days_since_last_success = NOW() - last_success_timestamp
    IF days_since_last_success > retention_window[concept]:
        mastery_score[concept] *= 0.85 # 15% decay
        append to weakness_state.decayed_concepts
→ Upsert mutated learning_dna to Qdrant
→ Trigger notification to student: "Time to revise Entropy!"

```

4. Enkrypt Safety Architecture

Mentra X — Enkrypt AI Safety Architecture

Purpose: Maximize Enkrypt AI Coverage scoring (20%)
SDK: Enkrypt AI Evaluation & Safety Framework

1. Why Enkrypt AI is Non-Negotiable

In competitive exam preparation, a single hallucinated formula can destroy a student's JEE rank. A fabricated historical date can cost them the UPSC Mains. A wrong chemical equation can eliminate their NEET score.

General LLMs hallucinate at rates of 3–15% depending on domain specificity. In a high-stakes educational context, even a 1% hallucination rate across 10 million queries produces 100,000 harmful responses.

Mentra X addresses this with **mandatory Enkrypt AI middleware** — positioned not as an optional quality layer, but as a hard blocker between the Mastra Tutor Agent and the student interface.

Enkrypt is not optional. It is the safety contract.

2. Architectural Position



3. Validation Pipelines

3.1 Mathematical Accuracy Validator

Trigger Condition: Response contains LaTeX, \$\$, =, integral signs, or numeric equations.

Process:

1. Extract all mathematical expressions from Tutor Agent output.
2. Parse into symbolic expression tree.
3. Verify: does step N logically follow from step N-1?
4. Spot-check final answer against known solution patterns.
5. Score: `math_accuracy (0.0-1.0)`.

Critical Exam Domains:

- JEE: Calculus, Coordinate Geometry, Differential Equations, Complex Numbers.
- CAT: Quantitative Aptitude, Data Interpretation.

Failure Action: Block output. Inject into Mastra context: "Your previous derivation contained a logical error in Step 3. Regenerate with a corrected intermediate step."

3.2 Science Fact Validator

Trigger Condition: Response tagged with `subject: Physics | Chemistry | Biology`.

Process:

1. Extract all factual claims (laws, constants, biological processes).
2. Cross-reference against embedded NCERT/standard textbook corpus in Qdrant (`reference_corpus` collection).
3. Score factual alignment: `science_accuracy (0.0-1.0)`.

Critical Exam Domains:

- NEET: Organic Chemistry, Human Physiology, Genetics.
- JEE: Thermodynamics, Electrostatics, Optics.

Failure Action:

- If fabricated constant detected (e.g., wrong value of Avogadro's number): **Hard block + fallback to safe textbook definition.**
 - If minor inaccuracy: **Warn with orange UI badge.**
-

3.3 Hallucination Detector

Trigger Condition: Response tagged with `subject: History | Polity | Economy | Geography` (UPSC/CAT).

Process:

1. Extract entity claims (names, dates, events, policy titles).
2. Embed claims.
3. Compare against known-good fact corpus via Qdrant semantic search.

4. Threshold: if no high-confidence match found for a claim, flag as potential hallucination.
5. Score: `hallucination_risk (0.0-1.0)`.

Failure Action:

- `hallucination_risk > 0.5`: Block + Regenerate.
 - `hallucination_risk 0.25-0.5`: Deliver with red caution badge: "This claim could not be verified. Cross-check with official sources."
-

3.4 Pedagogical Evaluator

Trigger Condition: Every single response (no exception).

Process:

1. Check: does the response explain the *why*, not just the *what*?
2. Check: does the response conclude with a guiding question or next step?
3. Check: is the explanation calibrated to the student's detected teaching level?
4. Score: `pedagogy_quality (0.0-1.0)`.

This evaluator does NOT block. It contributes to the overall confidence score and flags low-pedagogy responses for the Insight Agent to use in report generation.

3.5 Risk Classifier

Trigger Condition: All responses.

Process:

1. Scan for: harmful content, off-topic material, exam-rule violations (e.g., giving full exam answers).
2. Classify risk: `LOW | MEDIUM | HIGH`.

Failure Action:

- `HIGH` risk: Block entirely, log incident, trigger admin notification.
 - `MEDIUM`: Soft redirection ("That question falls outside the scope of this tutor").
-

3.6 Confidence Scorer (Aggregator)

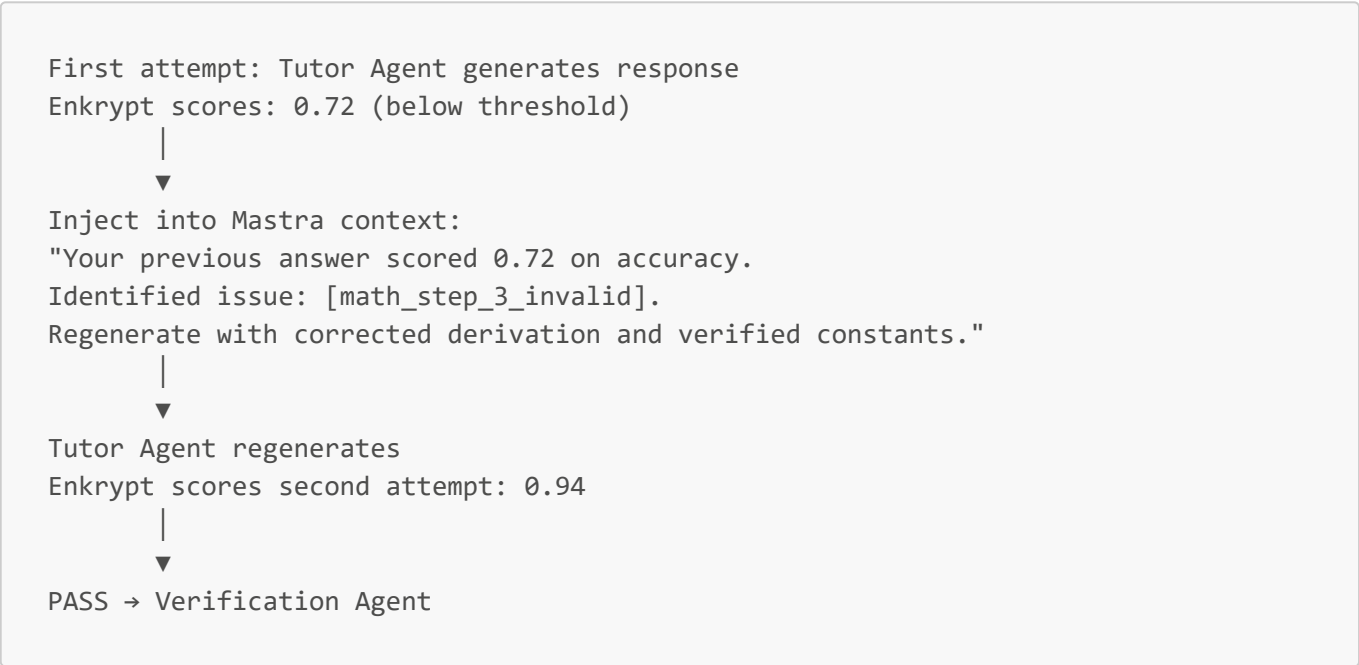
Aggregates all evaluator scores into a single `confidence_score (0.0-1.0)`:

```
confidence_score = (  
    0.35 × math_accuracy +  
    0.30 × science_accuracy +  
    0.20 × (1.0 - hallucination_risk) +  
    0.15 × pedagogy_quality  
)
```

Decision Matrix:

Confidence Score	Action	UI Indicator
≥ 0.90	Pass → Verification Agent	☑ No badge
0.75 – 0.89	Pass with warning	⦿ "AI generated — verify with textbook"
0.50 – 0.74	Block → Mastra regeneration (attempt 2)	⚠ Regenerating...
< 0.50	Hard block → Safe textbook fallback	⦿ "This concept exceeds AI confidence. Refer to Chapter X."

4. Regeneration Loop



Maximum regeneration attempts: 2.
If 2 failures: Hard fallback to safe pre-verified textbook excerpt.

5. Enkrypt Integration — Exact Code Hooks

```
from enkrypt_sdk import SafetyEvaluator

evaluator = SafetyEvaluator(domain="education", exam_type="JEE")

result = evaluator.evaluate(
    content=tutor_agent_output,
    subject_tag=request.subject,
    student_level=twin.behavior_state.preferred_level
)
```

```
if result.confidence_score >= 0.90:
    pass_to_verification_agent(result.content)
elif result.confidence_score >= 0.75:
    deliver_with_warning(result.content, badge="yellow")
else:
    trigger_mastra_regeneration(
        agent="tutor_agent",
        error_context=result.failure_reason
    )
```


5. Digital Twin Design

Mentra X — Student Digital Twin Design

Purpose: Primary Innovation Document

Status: The core differentiator of the Mentra X platform.

1. What is the Student Digital Twin?

The **Student Digital Twin** is a persistent, living computational model of a student's academic mind. Unlike a user profile (which stores what a student did), the Digital Twin models *how* they think, *why* they fail, and *what* they need next.

It is the first application of Digital Twin theory — pioneered in manufacturing by Siemens and NASA to model physical assets — to human cognitive science and personalized education.

Every student in Mentra X has exactly one Digital Twin. It is:

- **Initialized** on onboarding via the Assessment Agent.
 - **Mutated** after every interaction via the Verification Agent.
 - **Analyzed** every 5 sessions by the Weakness Intelligence Agent.
 - **Decayed** nightly to model natural forgetting.
 - **Retrieved** before every teaching session by the Memory Agent.
-

2. Twin Architecture: The Five States

State 1: Academic State

Tracks the student's formal position within structured curricula.

```
"academic_state": {  
  "exam_target": "JEE Advanced",  
  "enrolled_subjects": ["Physics", "Chemistry", "Mathematics"],  
  "current_chapters": {  
    "Physics": "Thermodynamics",  
    "Chemistry": "Organic Reactions",  
    "Mathematics": "Differential Equations"  
  },  
  "xp_total": 3450,  
  "streak_days": 21,  
  "completion_percentage": {  
    "Physics": 0.62,  
    "Chemistry": 0.45,  
    "Mathematics": 0.78  
  }  
}
```

Data Source: Mentra MySQL (Enrollments, XP, Course Progress)

Update Trigger: Course completion events, XP awards.

State 2: Knowledge State

The granular, concept-level mastery model. This is where individual academic intelligence lives.

```
"knowledge_state": {
  "Physics > Thermodynamics > Entropy": {
    "mastery_score": 0.38,
    "confidence_level": "low",
    "mistake_count": 14,
    "last_successful_recall": "2026-06-15T09:00:00Z",
    "total_doubt_sessions": 6,
    "explanation_levels_tried": [2, 3, 4],
    "resolved": false
  },
  "Mathematics > Differential Equations > Homogeneous": {
    "mastery_score": 0.89,
    "confidence_level": "high",
    "mistake_count": 1,
    "last_successful_recall": "2026-06-20T14:00:00Z",
    "resolved": true
  }
}
```

Data Source: Mastra Verification Agent (post-comprehension check), Quiz scores from MySQL.

Update Trigger: Every doubt session and micro-quiz result.

Mastery Score Range: 0.0 (no understanding) → 1.0 (expert).

Confidence Levels: **critical** < 0.30 | **low** 0.30–0.55 | **developing** 0.55–0.75 | **high** > 0.75

State 3: Skill State

Cross-cutting technical and cognitive skills beyond subject knowledge.

```
"skill_state": {
  "problem_solving": 0.71,
  "formula_recall": 0.44,
  "conceptual_understanding": 0.68,
  "exam_time_management": 0.52,
  "diagram_interpretation": 0.80
}
```

Data Source: Derived from Coding Platform scores, Quiz performance patterns, Interview results.

Update Trigger: Weekly aggregation from Mentra LMS analytics.

State 4: Learning DNA

The behavioral and cognitive fingerprint of the student. This is the highest-value state for teaching personalization.

```
"learning_dna": {
  "primary_learning_style": "Visual",
  "secondary_learning_style": "Narrative",
  "frustration_index": 0.61,
  "avg_session_duration_minutes": 28,
  "engagement_drop_minute": 25,
  "preferred_teaching_level": 3,
  "analogy_effectiveness": {
    "mechanical_analogies": 0.9,
    "mathematical_proofs": 0.3,
    "real_world_scenarios": 0.85
  },
  "response_to_correction": "receptive",
  "prefers_examples_before_rules": true,
  "works_well_under_time_pressure": false
}
```

Data Source: Mastra agent behavioral signals (how quickly does the student ask for simplification? How many times do they rephrase their doubt?).

Update Trigger: Every session. Exponential moving average smooths updates to prevent noise from single data points.

State 5: Weakness State

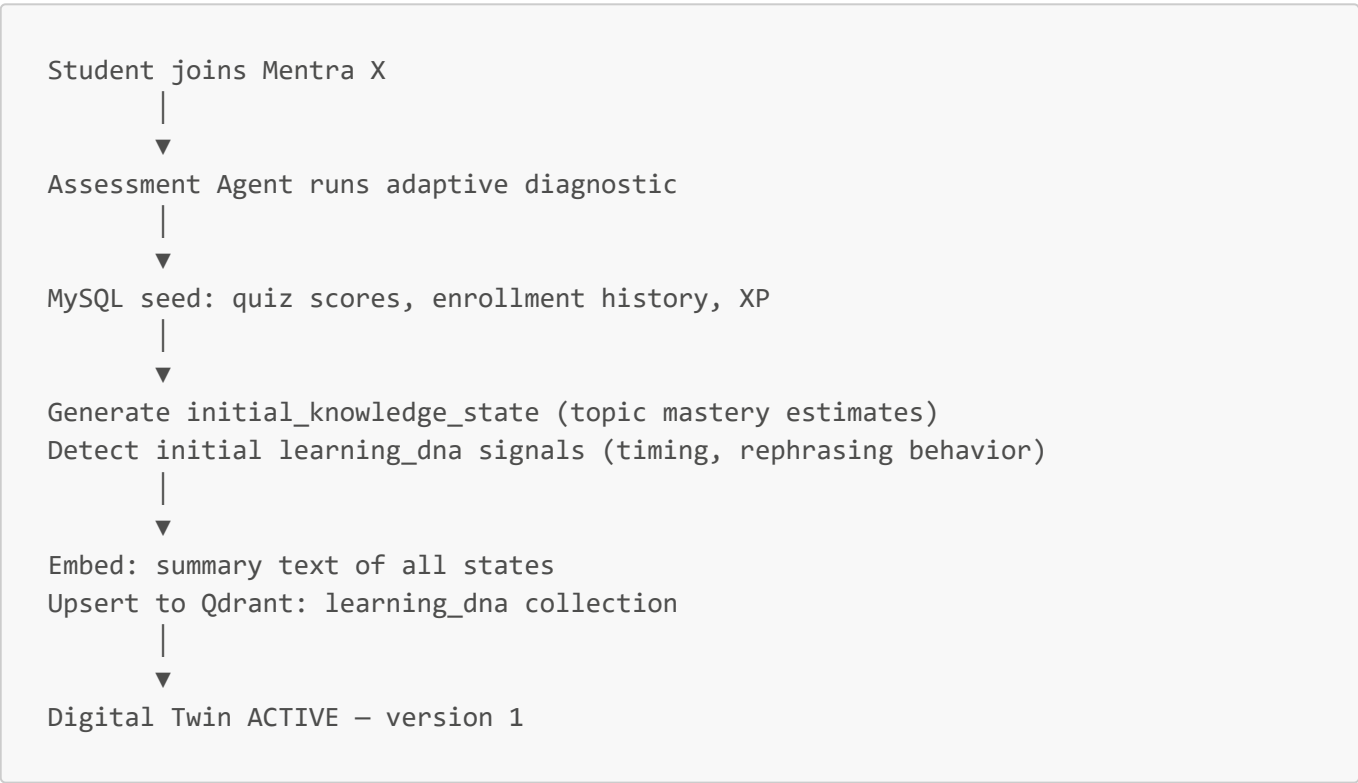
Synthesized by the Weakness Intelligence Agent from session logs.

```
"weakness_state": {
  "recurring_traps": [
    "Forgets negative sign when integrating sin(x)",
    "Confuses entropy with enthalpy"
  ],
  "low_mastery_concepts": [
    "Entropy",
    "Clausius Inequality",
    "Wave Optics Diffraction"
  ],
  "failure_patterns": [
    "Performs well on formula substitution, fails on conceptual derivation",
    "Answers correctly in isolation, fails under multi-step problems"
  ],
}
```

```
"decayed_concepts": [
  "Wave Optics (last review: 14 days ago)",
  "Electrostatics Gauss Law (last review: 21 days ago)"
],
"revision_urgency": {
  "Entropy": "CRITICAL",
  "Wave Optics": "HIGH",
  "Gauss Law": "MEDIUM"
}
}
```

Data Source: Weakness Intelligence Agent (Mastra cron), Qdrant semantic clustering.
Update Trigger: Every 5 sessions.

3. Twin Initialization Flow



4. Twin Mutation Protocol

Every mutation is versioned. Previous states are archived for trend analysis.

Event	Mutation
Student passes micro-quiz	mastery_score[topic] += 0.05, version++
Student fails micro-quiz	mastery_score[topic] -= 0.05, mistake_count++, version++
Student asks same question again	frustration_index += 0.03, version++

Event	Mutation
Session ends (5th session)	Full Weakness State rebuild, <code>version++</code>
7 days since last concept review	<code>mastery_score[topic] *= 0.85</code> (decay), <code>version++</code>
Student explicitly says "I understood"	<code>confidence_level</code> upgraded one tier

5. Personalization Logic

The Digital Twin directly controls the Tutor Agent's behavior through the following mappings:

Twin Signal	Agent Response
<code>frustration_index > 0.7</code>	Jump to Level 4 or 5 explanation immediately
<code>preferred_level = 3</code>	Always start at Level 3 for new concepts
<code>primary_style = Visual</code>	Prefer visual analogies, avoid formula-heavy proofs
<code>mastery_score < 0.3</code> (critical)	Trigger Weakness Agent early (3 sessions instead of 5)
<code>engagement_drop_minute = 25</code>	Insert a comprehension break at minute 22
<code>weakness_state.revision_urgency = CRITICAL</code>	Open session with "Before your doubt, let's briefly revise X"

6. Agent Workflows

Mentra X — Agent Workflows

Purpose: Detailed step-by-step workflows for all seven core interactions.

Workflow 1: Student Onboarding

```
graph TD
  A[New Student Registers] --> B[Flask Gateway: Create User in MySQL]
  B --> C[Assessment Agent Triggered]
  C --> D{Exam Target?}
  D -->|JEE| E[Load JEE Diagnostic Questions]
  D -->|NEET| F[Load NEET Diagnostic Questions]
  D -->|UPSC| G[Load UPSC Diagnostic Questions]
  D -->|CAT| H[Load CAT Diagnostic Questions]
  E & F & G & H --> I[Fetch MySQL Quiz History]
  I --> J[Generate Topic-Level Mastery Estimates]
  J --> K[Detect Behavioral Signals from Timing]
  K --> L[Initialize Learning DNA]
  L --> M[(Upsert to Qdrant: learning_dna)]
  M --> N[Student Enters Study Dashboard]
  N --> O[Display: Learning DNA Heatmap + First Recommendations]
```

Workflow 2: Adaptive Assessment

Trigger: New subject enrolled OR monthly calibration.

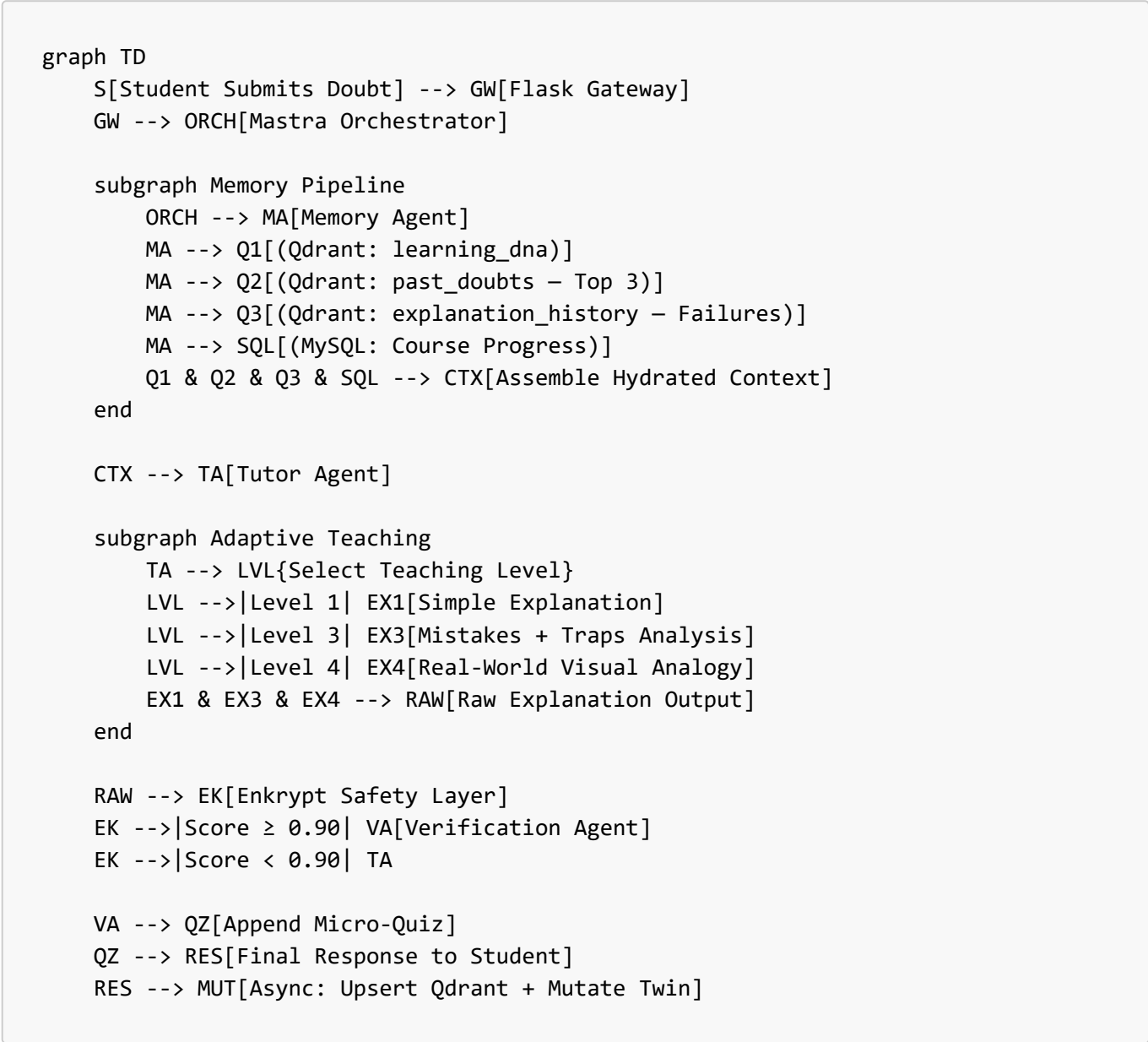
Step-by-Step:

Step	Actor	Action
1	Assessment Agent	Load topic list for exam target
2	Assessment Agent	Start with <code>difficulty = 0.5</code> (medium)
3	Student	Answers first question
4	Assessment Agent	Evaluate response via Mastra tool <code>evaluate_answer</code>
5a	Assessment Agent (if correct)	<code>difficulty += 0.1</code> , next topic same depth
5b	Assessment Agent (if wrong)	<code>difficulty -= 0.1</code> , record <code>failure(topic)</code>
6	Assessment Agent	Repeat steps 3-5 for all key topics
7	Assessment Agent	Synthesize <code>knowledge_state</code> object

Step	Actor	Action
8	Assessment Agent	Upsert to Qdrant <code>learning_dna</code>
9	Insight Agent	Generate "Your Baseline Report"

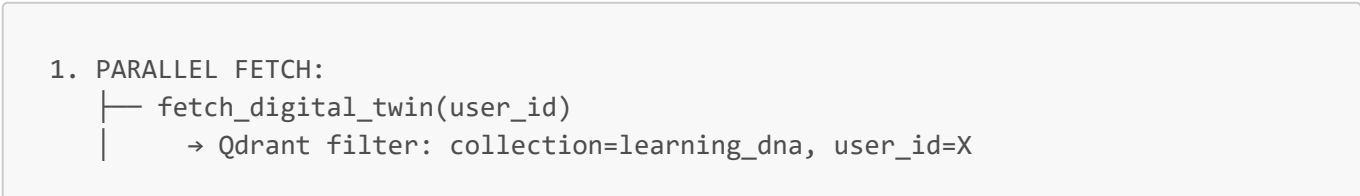
Workflow 3: Doubt Solving

Trigger: Student types a doubt in the Cognitive Chat UI.



Workflow 4: Memory Retrieval

Trigger: Called by Memory Agent at the start of every doubt session.



```

    → Returns: full twin JSON payload

— retrieve_past_doubts(embed(doubt_text), user_id)
    → Qdrant cosine search: collection=past_doubts, user_id=X
    → Top-3 by similarity
    → Returns: [{question_text, teaching_level_used, resolved}, ...]

— retrieve_failed_explanations(topic, user_id)
    → Qdrant filter: collection=explanation_history
      user_id=X, concept=topic, student_success_flag=False
    → Returns: [teaching_level: 2, analogy: "Carnot cycle example"], ...

— fetch_mysql_context(user_id)
    → Course progress, current chapter, XP streak

```

2. ASSEMBLE:

```

hydrated_context = {
  digital_twin: ...,
  similar_prior_doubts: [...],
  failed_approaches: [...],
  mysql_progress: {...}
}

```

3. PASS to Tutor Agent as system prompt extension

Workflow 5: Understanding Verification

Trigger: Tutor Agent output has passed Enkrypt validation.

1. Verification Agent receives:
 - explanation_text (Enkrypt-validated)
 - concepts_addressed: ["entropy", "second law"]
 - teaching_level_used: 3
 - twin.learning_style: "Visual"
2. Generate micro-quiz question:
 - "If the temperature of a heat reservoir increases,
what happens to the entropy change of the universe?"
 - (Calibrated to NOT be identical to the doubt – tests transfer of understanding)
3. Student answers
4. Evaluate answer:
 - Semantic similarity to correct answer
 - Confidence threshold: 0.75 = PASS
- 5a. IF PASS:
 - "Great! You understood the core concept."
 - Upsert: explanation_history (success_flag=True)
 - Mutate: learning_dna mastery_score[entropy] += 0.05

5b. IF FAIL:

- "That's not quite right. Let me try a different angle."
- Auto-trigger Tutor Agent at Level (current_level + 1)
- Upsert: explanation_history (success_flag=False)
- Mutate: learning_dna mistake_count[entropy] += 1

Workflow 6: Weak Area Detection

Trigger: Mastra cron-workflow fires after every 5th completed session.

1. Fetch last 5 session_logs from Qdrant (filter: user_id=X, last 5)
2. Extract all failed_concepts from session payloads:
 - ["entropy", "Clausius inequality", "Carnot efficiency", "entropy", "entropy", "heat engine efficiency"]
3. Cluster by semantic similarity:
 - Cluster 1: ["entropy", "Clausius inequality", "entropy"] → macro: "Second Law & Entropy"
 - Cluster 2: ["Carnot efficiency", "heat engine efficiency"] → macro: "Carnot Cycle Applications"
4. Rank clusters:
 - Second Law & Entropy: frequency=3, severity=HIGH
 - Carnot Cycle: frequency=2, severity=MEDIUM
5. Upsert to Qdrant weak_concepts:
 - {macro_weakness: "Second Law & Entropy", severity_score: 0.82, ...}
6. Mutate learning_dna:
 - weakness_state.low_mastery_concepts += ["Entropy"]
 - weakness_state.revision_urgency["Entropy"] = "CRITICAL"
7. Generate revision_plan:
 - "Day 1: Re-read Entropy definition (Level 4 analogy)
 - Day 2: Solve Clausius inequality problems
 - Day 3: Full Carnot cycle mock problem set"
8. Push revision_plan to MySQL student dashboard

Workflow 7: Progress Reporting

Trigger: Weekly (every 7 days) OR student requests "Show my report".

1. Insight Agent fetches from:
 - └─ Qdrant learning_dna: current vs 7-days-ago mastery delta

- └─ Qdrant weak_concepts: active weaknesses
- └─ MySQL: XP earned, streak, sessions completed

2. Synthesize weekly report:

 WEEKLY LEARNING REPORT – JEE Advanced

Sessions this week: 14

XP Earned: 340

Concepts Mastered: 3 (Kinematics, Vectors, Integration basics)

⚠ CRITICAL: Entropy (0 sessions this week – mastery decayed to 0.32)

⚠ HIGH: Wave Optics (last revised 9 days ago)

🕒 Next 3 Days:

→ Day 1: Entropy – Level 4 Visual Session

→ Day 2: Wave Optics – Practice Problem Set

→ Day 3: Full Mixed Mock (Physics)

3. Deliver via Chat UI + Email notification

4. Archive in MySQL for long-term trend analysis

Built by Sujith Kumar AI HiDevs × Mastra Hackathon 2026